Pretrain a Remote Sensing Foundation Model by Promoting Intra-Instance Similarity

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***Abstract*— Self-supervised learning (SSL) has gained significant traction within the remote sensing community, with pretraining a foundation model on large-scale unlabeled datasets for the interpretation of remote sensing images (RSIs) emerging as a trending direction. This approach aims to supplant the conventional practice of loading ImageNet pretrained weights, offering a more versatile and potentially more effective solution for handling RSIs. Among SSL techniques, contrastive learning excels in extracting general representations in the field of remote sensing. However, its excessive focus on inter-instance discrimination hinders the effectiveness of pretraining due to the diverse and complex geographical information present in RSIs. Moreover, the typical *two-variations-as-one-pair* pattern may be suboptimal, particularly given the temporal information specific to RSIs. In this article, we propose a novel method called promoting intra-instance similarity (PIS) for short, which leverages the temporal information specific to RSIs and increases the intra-instance variations to expand the positive representation space. Additionally, by PIS within this space, our foundation models develop the ability to extract more general and instance-invariant features that prove beneficial for various downstream tasks. Experiments show that our PIS method achieves state-of-the-art (SOTA) performance on ten datasets across four downstream remote sensing tasks, demonstrating the generalizability and efficacy of the proposed method. Through our preliminary investigation into intra-instance characteristics, we believe there exists substantial potential in this aspect, holding considerable promise for further exploration. The codes are available on the website: https://github.com/ShawnAn-WHU/PIS.git.**

***Index Terms*— Foundation model, intra-instance similarity, positive representation space, remote sensing, self-supervised learning (SSL).**

I. Introduction

THE interpretation for remote sensing images (RSIs) has been thriving thanks to the explosive increase in satellite launches and the prosperity of deep learning methods in the field of remote sensing. Furthermore, these advancements have contributed to a prominent rise in numerous remote sensing downstream tasks compared to the conventional methods, including land-cover mapping [1], [2], hyperspectral image classification [3], [4], [5], [6], [7], and urban traffic surveillance [8]. As a result, researchers and decision-makers can gain more data support to meet the demands for accurate, timely and feasible insights into environment management [9], disaster monitoring, and postdisaster recovery efforts. However, despite the increase of satellites launched, as well as the number of RSIs captured, the gap between these images and their detailed annotations has not been bridged in the context of conventional supervised learning [10]. Manually annotating large-scale datasets is an extremely cumbersome, time-consuming, and expensive procedure, especially for RSIs with intricated geographic feature information, which significantly impedes the further development of advanced remote sensing applications. Meanwhile, in the case of RSIs, a type of data that perceives the physical world, the useful information it contains can rapidly fall out of date due to human activities, such as deforestation [11] and building construction [12], and natural phenomena, including climate change [13] and earthquakes [14]. This dynamic nature of RSIs not only adds another layer of complexity to the annotating process, but also necessitates specialized knowledge for accurate interpretation, thereby further amplifying the challenges involved. Due to the scarcity of high-quality RSI labels for downstream tasks, the prevalent approach to interpreting RSIs is to initially load ImageNet pretrained weights, followed by employing remote sensing data and task-specific label information at hand for model training. Although this strategy effectively accelerates the convergence speed and enhances the performance of specific tasks to a certain degree, it also raises several concerns that need to be addressed.

1) *Domain Gap Between RSIs and Natural Images:* While the natural images (e.g., ImageNet [15]) are predominantly object-centric, RSIs are captured from a bird-eye view, where the scale and orientation of the objects, such as aircraft at the airport shown in Fig. 1(a), may exhibit significant variations. Additionally, RSIs typically cover a large spatial area, and therefore, often contain numerous small and dense objects, e.g., cars in the parking lot depicted in Fig. 1(b). Beyond the primary objects of interest, such as buildings shown in Fig. 1(c), RSIs also feature complex background information and are

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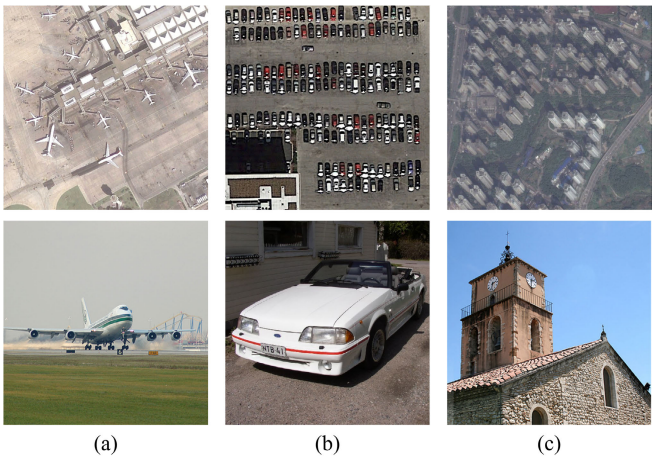
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(a) (b) (c)

Fig. 1. Examples of comparisons between (top) RSIs and (bottom) natural images. (a) Aircraft. (b) Cars. (c) Buildings.

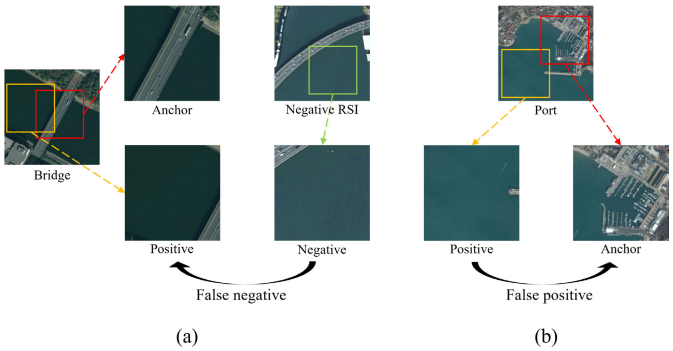


Fig. 2. Examples of the false-negative and false-positive sample. The negative sample from the negative RSI is so similar to the positive image in (a) that it could be a false negative one corresponding to the anchor of the Bridge; the positive sample in (b) is semantically different from the anchor of the Port, which could be seen as a false-positive sample.

subject to imaging interference, which is affected by environmental factors like weather, lighting, and clouds.

2) *Weak Generalization Capacity for Various Scenes:* There are only a small number of off-the-shelf high-quality annotations, which typically concentrate on specific geographic areas or types of landscapes. However, RSIs inherently display substantial heterogeneity, encompassing diverse and complex terrains and environments. This wide geographic and environmental variety implies that no single dataset can fully represent the data distribution of all scenes and objects. Consequently, the performance of the model would deteriorate when applied to cross-domain tasks. This challenge in generalization is particularly pronounced in the interpretation of RSIs due to the distinct cross-regional and cross-environmental characteristics inherent to this type of image.

3) *Inefficiency of Practical Remote Sensing Applications:* For different vision tasks (e.g., classification and segmentation), it is necessary to annotate RSIs separately to acquire specific formats of labels. Given the extensive coverage and complex geographical information of RSIs, annotation work requires not only just high precision but also specialized knowledge in areas such as geography and environmental science. Additionally, various models and configurations (e.g., loss functions and hyperparameters) are required for different tasks, thus retraining from scratch is inevitable, which dramatically diminishes the efficiency of practical applications. Furthermore, the increasingly huge volume of RSIs that can be handily accessed remains untapped.

In summary, the current paradigm of interpretation for RSIs appears to be less suitable in recent contexts, highlighting the urgent need for a universal approach that fully leverages the vast volume of RSIs without any reliance on manual annotations.

In recent years, self-supervised learning (SSL) has gained great popularity within the remote sensing community and designing a foundation model with SSL to extract more representative features of RSIs is inclined to be a know-it-all for various remote sensing downstream tasks [16]. Based on a task-agnostic objective, referred to as a pretext task, SSL utilizes a large-scale unlabeled dataset to train a model by optimizing this specific objective without any manual annotation. Through this hand-crafted pretext task, the model is trained to extract high-level representations of the input RSIs, and then can be transferred to supervised downstream tasks where only a small number of annotations are available.

Contrastive learning, one of the mainstreams of SSL, encourages the resemblance between semantically similar inputs (e.g., two augmented variations of the same image) while preventing model collapse with the technique of negative sampling [17], [18], [19], [20], [21], clustering [22], [23], knowledge distillation [24], [25], [26], [27], or redundancy reduction [28], [29], [30], further qualifying the model to excavate the latent representative features during pretraining. Such contrastive learning methods with negative sampling, akin to a form of classification, aim to pull positive samples closer together in the representation space, while simultaneously pushing negative samples apart to avoid trivial solutions. However, it is precisely because of the introduction of negative samples that either large batch size (e.g., 4096 or 8192 in [18]) to increase the number of negative samples or a momentum encoder [19], [20] to dynamically update the features of negative samples is essential in order to incorporate richer visual information. Furthermore, these methods overly concentrate on inter-instance discrimination, where the inevitable false negative samples may lead to a noisy training objective [31]. Owing to the diversity and complexity of geographical information contained in RSIs, this challenge is especially pronounced in the field of remote sensing. For example, the augmented variation of the negative RSI shown in Fig. 2(a) is similar to the positive sample of the corresponding anchor image. Therefore, expanding the positive representation space to alleviate the impact of false negative samples is crucially important. Clustering-based contrastive learning methods learn data representation by using clustering algorithms to group similar features together in the embedding space. DeepCluster [22] proposed to leverage K-means clustering of representations from previous iterations as pseudo-labels for new

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representations. This process necessitates a costly and asynchronously conducted clustering phase, which complicates scaling the method. SwAV [23] innovated by incorporating online clustering into this kind of method. It posits that different views of the same images should correspond to the same (clustered) prototypes. Meanwhile, an online computing strategy under a balanced partition constraint for each batch, which is solved by the Sinkhorn–Knopp transform [32], was designed to accelerate code (cluster assignment) computing. These clustering approaches can be viewed as contrastive learning at the cluster level, which still requires numerous negative comparisons to work well, suffering from the impact of false-negative samples. Another type of contrastive learning that integrates knowledge distillation typically employs a teacher–student architecture and optimizes a similarity metric of two augmented variations from the same image. This pattern of treating two variations as one single pair might be suboptimal, as limited variations per image can restrict intra-instance diversity, and the possible false-positive sample introduces significant noise into the training process [31] shown in Fig. 2(b). However, we cannot simply discard the false-positive sample, because it is part of the original image. Contrastive learning methods that utilize redundancy reduction as a way to avoid model collapse share the same challenges as above. Therefore, it is worth generalizing as many potential variations for each image as possible within one single process to mitigate the influence introduced by false positive samples.

Due to the inherent shortcomings of the conventional contrastive learning framework, foundation models pretrained by these SSL methods do not perform very well. Generalizing more potential intra-instance variations, i.e., expanding the instance representation spaces, appears to be a promising solution to address the aforementioned challenges. And intuitively the intra-instance characteristics within the expanded RSIs representation spaces, which depart from the previous framework, can be further explored.

In this article, we propose a remote sensing foundation model emphasizing promoting intra-instance similarity (PIS), which capitalizes on the distinct attributes of RSIs and harnesses a broader range of potential variations to unearth the intra-instance characteristics, thereby extracting more resilient representations for downstream tasks. A comprehensive remote sensing dataset encompassing a wide variety of scenes and objects serves as a foundational element for SSL pretraining. Based on the SSL4EO-S12 dataset [33] which consists of 251 079 locations and four seasonal timestamps per location, we employ PIS to carry out pretraining through the following pipeline. Initially, we apply multiple artificial data augmentations to each seasonal timestamp of one selected location to create an instance subspace. Subsequently, a feature extractor consisting of a standard backbone network (ResNet [34] or Swin Transformer [35]) along with several MLP layers is utilized to derive latent representations for the augmented variations of the instance, which are dispersed across the representation space. Following this, the average representation of these variations, designated as centroid, is calculated. Throughout the pretraining process, all the scattered representations are guided to converge toward this centroid. Concurrently, a covariance regularization is applied to prevent model collapse. Upon completing the pretraining and acquiring the foundation model by PIS, the latent representations extracted by the pretrained model become capable of supporting a variety of downstream tasks. Experimental results on four typical remote sensing downstream tasks (scene classification, semantic segmentation, change detection, and object detection) demonstrate that our two versions of remote sensing foundation model can achieve state-of-the-art (SOTA) performance on ten publicly available datasets. Furthermore, ablation studies suggest that enriching the positive representation space by seasonal pairs and increasing the number of variations for each instance is worth boosting the robustness of the latent representations.

In summary, the contributions of this article are four folds.

1) Given the limitations of the general paradigm that loads ImageNet pretrained models, we pretrain two versions of remote sensing foundation model to enhance the efficiency and effectiveness in practical applications.
2) Considering the excessive attention to inter-instance discrimination and the suboptimal *two-variations-as-one-pair* pattern, we propose to expand the positive representation space followed by exploring the intra-instance characteristics to alleviate the impact of false-negative and false-positive samples during pretraining.
3) By aggregating the scattered representations within the representation space toward their centroid, i.e., promoting the similarity between each variation's representation and centroid, the model can extract more robust features to recognize different variations from the same instance, especially when the covariance regularization aids in preventing trivial solutions. Therefore, the proposed PIS method enables the model to learn high-level representations during the pretraining process, making it versatile to a broad spectrum of downstream tasks.
4) We fine-tune the two versions of the proposed pretrained model on four remote sensing downstream tasks. The experimental results show that our PIS method achieves SOTA performance across ten datasets, confirming the effectiveness and generalization capability of our PIS foundation models on practical applications.

The remainder of the article is organized as follows. Section II provides a review of related works. The details of the proposed pretraining method PIS will be described in Section III. The experimental results on four downstream tasks and the corresponding ablation studies are presented in Section IV. Finally, Section V concludes this article.

II. Related Works

In this section, we primarily review the relevant works about remote sensing foundation models and the methods to exploit the intra-instance characteristics.

A. SSL and Foundation Models in Remote Sensing

Masked autoencoder (MAE) [36] introduced a novel technique of reconstructing the partially masked image based on

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the visible patches to the realm of SSL methods. Subsequently, several research initiatives have adopted this approach to enhance the diversity of the foundation models in remote sensing. For large-scale and arbitrarily oriented objects in RSIs, Wang et al. [37] designed a new rotated varied-size attention (RVSA) mechanism, which significantly reduces computational costs and memory usage. Combined with MAE pretraining, they made the first attempt to pretrain plain ViTs with about 100 million parameters on RSIs. To address the issue where small targets in RSIs might be completely obscured by random masks, RingMo [38] introduced a PIMask strategy within image blocks to preserve the feature information of small objects. However, this inadvertently increased the computational costs and memory usage of the pretraining procedure. On another front, SatMAE [39] proposed a novel method to leverage temporal or multispectral information in RSIs to improve SSL with MAE. This method included a temporal embedding along with independently masking image patches across time to leverage temporal dynamics. It also demonstrated the benefits of encoding multispectral data as groups of bands with distinct spectral positional encodings.

As contrastive learning continues to gain an enduring popularity within the computer vision community [17], [18], [24], [25], [26], its potential is also increasingly being explored in the field of remote sensing. Based on a large-scale dataset containing a wide array of aerial scenes and other additional information specific to remote sensing, the pretrained models or foundation models are adept at mining representative latent features, upon which typical downstream tasks, such as scene classification, semantic segmentation, and change detection, experience substantial improvement when fine-tuning using a limited quantity of available labeled data.

Tile2Vec [40] is one of the earliest self-supervised works that utilize contrastive learning for interpreting RSIs. The researchers developed a convolutional neural network (CNN) that learns from triplets of RSI tiles. Each triplet is composed of an anchor tile (t_a), a geographically close neighboring tile (t_n), and a geographically distant tile (t_d), among which a triplet loss is optimized in order to pull neighboring tiles closer and push distant tiles far away in the representation space. Following the design in SimCLR [18], Hou et al. [41] proposed to leverage contrastive learning for hyperspectral imagery classification based on a single scene, thereby diminishing the need for dense annotations. Considering extra information specific to RSIs, Manas et al. [20] first gathered a large-scale, unlabeled remote sensing dataset containing RSIs from multiple locations at different timestamps. Regarding the temporal information in seasonal pairs from the same location at different points of time as a natural supplementary for artificial augmentation, seasonal contrast (SeCo), adopting the framework in MoCo [19], projects the common representations into several embedding subspaces to learn better visual representations that simultaneously variant and invariant to seasonal changes. Scheibenreif et al. [21] introduced Swin Transformer [35] into SimCLR [18], and proposed a dual-stream network to optimize the contrastive loss between paired optical Sentinel-2 and SAR Sentinel-1 RSIs, thereby fine-tuning with as little as 10% of labeled data can outperform fully supervised approaches.

Recently, Tao et al. [42] claimed that relying solely on RSIs could restrict the discrimination and generalizability of the learned features, attributing to lower spatial resolution and nonobject centralization of RSIs. Following a human-like SSL learning mechanism, they introduced the original vision (TOV) model, which first learns general knowledge from web-scale natural images and then focuses on domain-relevant knowledge from unlabeled RSIs. Considering the redundancy of such a multistage strategy, GeRSP, which employs a teacher–student architecture, was proposed by Huang et al. [43] to simultaneously learn representations with general and special knowledge.

It is noteworthy that the selection of positive and negative samples in these methods are all based on simple spatial and spectral augmentations (e.g., random cropping, color Jittering, and Gaussian Blur). As previously mentioned, these augmentations are judiciously chosen because these methods do not fare well under heavy augmentation (e.g., Jigsaw) [44]. Additionally, the false-negative and false-positive samples usually result in a lousy performance. The proposed PIS, which aggregates more potential variations scattered throughout the expanded positive representation space toward their centroid, presents an effective solution to these challenges.

B. Explorations Into Intra-Instance Characteristics

To tackle the challenges stemming from over attention to inter-instance discrimination and the occurrence of false-negative/positive samples, many efforts have been made that can be regarded as expanding the positive representation space, namely, explorations into intra-instance characteristics.

Different from previous contrastive learning methods with knowledge distillation, Caron et al. [26] constructed two global and several local variations with multicrop strategy [23] and passed them through the teacher and student network, respectively to encourage "local-to-global" correspondences. Xu et al. [45] combined the inter-instance discrimination and intra-instance variations and introduced KSCL to improve conventional contrastive learning methods by projecting the query onto the associated subspaces formed by each instance and assigning the query to the instance class with the shortest projection distance. Ci et al. [46] proposed Fast-MoCo, which follows a divide-encode-combine and then contrast pipeline. This approach utilizes combinatorial patches to construct multiple positive pairs, thereby offering ample supervision signals.

Although generating more local crops or variations indeed expands the representation spaces, these methods remain bound by the conventional paradigm of contrastive learning with negative sampling or knowledge distillation. And only characteristics between the positive and negative spaces are explored. In contrast, our proposed method delves into the inner nature within the positive representation space itself by PIS, thereby exploring the intra-instance characteristics more deeply.

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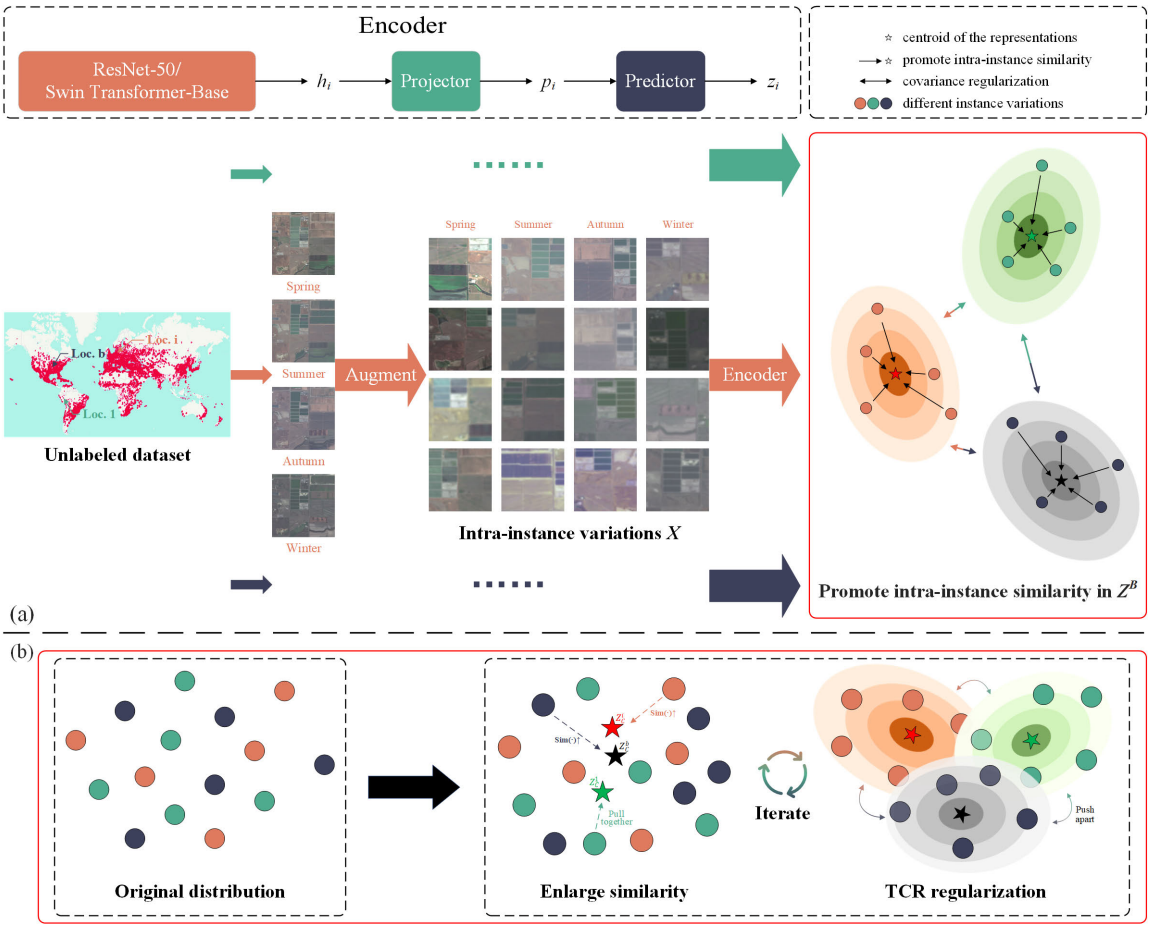


Fig. 3. (a) Overall framework of the proposed PIS method. (b) Details in the batch of intra-instance spaces Z^B during training process. The main idea is to pull intra-instance variations together within each positive representation space by increasing the similarity between z_k^i and the centroid Z_C^i. Meanwhile, TCR covariance regularization is employed to push different intra-instance spaces in Z^B apart, thereby preventing model collapse.

III. Methodology

Given a large-scale unlabeled dataset of RSIs with seasonal information, the proposed PIS method learns representations through a modified contrastive learning framework. In this framework, the positive representation space of each instance is expanded, and intra-instance similarity is enhanced within that space, thereby facilitating a more robust understanding of the data's pattern and variations to spatial and seasonal changes. For each of the four timestamps in one location, we generate a diverse range of intra-instance variations by artificial augmentations, yielding both artificially induced and temporal variations for each location, i.e., instance. Similar to preceding works, PIS incorporates an encoder, consisting of a backbone, a projector, and a predictor, to extract latent representations of these variations, which are then aggregated toward their centroid in order to force invariance to artificial and seasonal changes. Meanwhile, a covariance regularization term is adopted to prevent model collapse. To enhance the generalizability and adaptability of our pretrained model, we select two classical backbones: ResNet-50 [34] and its parameter-equivalent counterpart, Swin Transformer-Base [35]. The proposed PIS method is depicted in Fig. 3. In Sections III-A and III-B, we will delve deeper into each component, providing more detailed insights.

A. Pretraining Dataset and Intra-Instance Variations

A well-constructed pretraining dataset lays a solid foundation for SSL. The SSL4EO-S12 dataset [33] is a large-scale unlabeled dataset with multitemporal and multimodal RSIs collected by the European Space Agency's Sentinel mission. It contains about one million image triplets (Sentinel-1 dual-pol SAR, Sentinel-2 top-of-atmosphere multispectral, Sentinel-2 surface reflectance multispectral) from 251 079 locations across the globe, and four seasonal timestamps are included for each location. SSL4EO-S12 is collected by Google Earth Engine [47] and the image size for each patch is 264 × 264 with a resolution of 10 m, covering an area of 2640 × 2640 m.

For each instance, the four seasonal timestamps are ready-made and natural intra-instance variations due to the temporal information specific to RSIs [20]. However, the seasonal instances alone are not enough for expanding the positive representation space. To generalize more intra-instance variations, we further apply artificial augmentations, including random crop and resize, random horizontal flip, grayscale, color jittering, Gaussian blur, and solarization, to create $(N/4)$ views for each timestamp, resulting in a total of N augmented variations. The N variations form a unique representation space, which stands for

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the diversified inner nature of the corresponding instance itself.

B. Promote Intra-Instance Similarity

Several prior works have demonstrated that simply averaging the embeddings of cropped patches works reasonably well to represent whole-image embeddings [46], [48], [49]. Within the instance representation space and subject to a covariance regularization constraint, we promote the intra-instance similarity by aligning the representation of each variation closer to the space centroid, which is derived by averaging the representations of all the intra-instance variations.

As shown in Fig. 3, given intra-instance variations $X = [x_1, \ldots, x_N]$, we get embeddings $H = [h_1, \ldots, h_N] = f(X; \theta)$ and $P = [p_1, \ldots, p_N] = \phi(H)$, where $F(\cdot; \theta)$ is the backbone network (ResNet-50 or Swin Transformer-Base) parameterized by θ and the projector $\phi(\cdot)$ is a MLP network with only one fully connected layer. To further improve the representations learned, we project and l_2-normalize P through the predictor $\psi(\cdot)$ with two fully connected layers to get the representation space $Z = [z_1, \ldots, z_N] = \psi(P)$. Note that the predictor only influences the pretraining stage; the downstream classification task does not use it. For a batch of b RSIs during pretraining, we get $Z^B = [Z^1, \ldots, Z^b]$ following the above steps, where Z^i denotes the ith intra-instance representation space. For each Z^i, the centroid is determined by averaging all the elements inside as follows:

$$Z_C^i = \frac{1}{N} \sum_{k=1}^{N} z_k^i \tag{1}$$

where Z_C^i and z_k^i denoted the centroid of the ith representation space and the kth representation in this space, respectively. Then, we achieve the goal of PIS from a batch perspective by a similarity loss S following:

$$S\left(Z^B\right) = \frac{1}{b} \sum_{i=1}^{b} \cdot \frac{1}{N} \sum_{k=1}^{N} \operatorname{sim}\left(z_k^i, Z_C^i\right) \tag{2}$$

where $\operatorname{sim}(\cdot)$ represents the cosine similarity metric.

Only PIS may result in a trivial solution (e.g., identity mapping). Therefore, we adopt total coding rate (TCR) [50], [51] as a covariance regularization constraint to prevent the intra-instance representation spaces from collapsing to a singular point. More specifically, view Z^B as $T = [T_1, \ldots, T_N]$ where $T_k = [z_k^1, \ldots, z_k^b]$, TCR is calculated as follows:

$$R\left(Z^B\right) = \frac{1}{N} \sum_{k=1}^{N} \frac{1}{2} \log\det\left(I + \frac{d}{b\epsilon^2} T_k T_k^\top\right) \tag{3}$$

where d is the dimension of z_k^i, and $\epsilon > 0$ is a distortion coefficient. This constraint achieves a similar outcome to the covariance regularization term in VICReg [29] and Barlow Twins [28], whose optimization target is to make the covariance matrix of latent vector (T_k) approach the diagonal matrix. This ensures that the extracted features are distinct across different instances, thereby halting the model's inclination toward a shortcut path.

Overall, we want to promote intra-instance similarity by enlarging $S(Z^B)$ while keeping TCR loss nonattenuating. To sum up, the final training objective can be formulated as follows:

$$\max \mathcal{L} = \alpha S\left(Z^B\right) + \beta R\left(Z^B\right) \tag{4}$$

where α and β represent the weights of similarity loss and TCR loss, respectively.

IV. Experiments

In this work, we leverage the PIS method to train remote sensing foundation models with the RGB-version SSL4EO-S12 dataset [33]. For simplicity, our two foundation models are referred to as PIS-R50 and PIS-SwinB in subsequent discussions according to the backbone used in their encoder (R50 for Resnet-50 and SwinB for Swin Transformer-Base). To evaluate the performance of the pretrained models, we finetune them with a small amount of labeled data across ten downstream datasets for four remote sensing tasks: scene classification, semantic segmentation, change detection, and object detection. Experimental results highlight the SOTA performance on typical downstream tasks, even under scenarios with limited labels. Furthermore, the ablation studies on seasonal pairs and the number of intra-instance variations reveal that the proposed method is effective at boosting the robustness of learned representations. To give a qualitative analysis, we also visualize the learned features of PIS and comparison models by using t-SNE [52].

Pretraining Implementation Details: All experiments for pretraining are implemented by the Pytorch [53] framework with 6 NVIDIA GeForce RTX 3090 GPUs. To facilitate straightforward comparisons with other methods, we adopt the compressed 8-bit RGB version of SSL4EO-S12 dataset, where the raw 16/32-bit values are normalized by mean and standard deviation, and converted to unit 8, for our pretraining implementation. All input variations are randomly cropped to cover a 25% area of the original RSIs, and resized into 224×224 at the first step in the Augment stage.

Encoder Architecture: For the adaptability of the pretrained models, we adopt the widely used ResNet-50 [34] and Swin Transformer-Base [35] as the backbone networks of the encoder, with output dimensions being 2048 and 1024, respectively. The projector is designed with a fully connected layer, succeeded by batch normalization [54] and rectified linear units (ReLU) [55], featuring an output dimension of 4096. Meanwhile, the predictor comprises a linear layer, also outputting at 4096 dimensions, followed by batch normalization [54], ReLU [55], a second linear layer reducing to an output dimension of 1024 and a l_2-normalization layer.

Hyperparameter and Pretraining Settings: We set the number of intra-instance variations $N = 16$, i.e., four for each timestamp. For the ResNet-50 version, the SGD optimizer is used, featuring a momentum of 0.9 and a weight decay of $1e^{-4}$. Within the loss function, the weights α and β are set to 400 and 1, respectively. For the Swin Transformer-Base version, we opt for the AdamW optimizer with a weight decay of 0.01. The hyperparameters of the optimizer are set to

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TABLE I
DATASETS FOR SCENE CLASSIFICATION EXPERIMENTS

Dataset	UC Merced	AID	PatternNet	NR	EuroSAT
Number of categories	21	30	38	45	10
Number of samples	2,100	10,000	30,400	31,500	27,000
Spatial resolution (m)	0.3	$0.5 \sim 8$	$0.062 \sim 4.693$	$0.2 \sim 30$	$10 \sim 60$
Image sizes	256×256	600×600	256×256	256×256	64×64

$\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\epsilon = 1e^{-8}$. The corresponding weights in the loss function are adjusted to $\alpha = 200$ and $\beta = 5$. Both versions undergo a pretraining phase for 30 epochs with a batch size of 128 for ResNet-50 and 48 for Swin Transformer-Base version. The initial learning rates are tailored to 0.3 and $3e^{-4}$, respectively, and follow a cosine annealing schedule with a final value of 0. In the TCR loss, ϵ^2 is fixed to 0.2.

Comparison Methods: In the experiments, we compare our two versions of foundation model against eight other related works, including two basic model initialization methods, one contrastive learning method based on clustering, two SSL methods based on MoCo [19], two recently proposed remote sensing foundation models with the assistance of web-scale natural images, and one generative method, MAE [36].

1) *Random-Initialized ResNet-50:* The ResNet-50 [34] backbone is randomly initialized, i.e., the encoder is trained from scratch for each remote sensing downstream task.
2) *ImageNet-Pretrained ResNet-50:* An off-the-shelf network weight pretrained by ImageNet [15] is loaded into the backbone of encoder for each downstream task, which is the general paradigm for most methods in the field of remote sensing.
3) *SwAV [23]:* SwAV incorporates clustering into a Siamese network by computing the assignment from one view and predicting it from another. An online clustering strategy is performed under a balanced partition constraint for each batch. The model is pretrained on ImageNet [15] using ResNet-50 [34].
4) *SeCo [20]:* SeCo is a remote sensing foundation model pretrained on a self-constructed dataset with one million RSIs from different seasons. Equipped with temporal information and multiple embedding subspaces, it follows a framework akin to MoCo [19] and takes advantage of time and position invariance to learn transferable representations for downstream tasks.
5) *SSL4EO-S12 [33]:* Besides a large-scale dataset, SSL4EO-S12 also provides a variety of pretrained models, among which we choose the MoCo-pretrained ResNet-50 model based on the RGB-version dataset for comparison.
6) *TOV [42]:* TOV adopts ResNet-50 and is pretrained using TOV-NI dataset (one million natural images) and TOV-RS dataset (0.5 million RSIs), continually along a human-like learning path.
7) *GeRSP [43]:* GeRSP contains two pretraining branches, an SSL branch for unlabeled RSIs and a supervised branch for labeled natural images. These two branches are combined by a teacher–student architecture to simultaneously learn representations with general and specific knowledge, thereby generating a powerful foundation model for downstream tasks.
8) *MAE [36]:* MAE is one of the generative pretraining methods in contrast to the contrastive learning. It adopts an asymmetric encoder–decoder architecture, with an encoder that operates only on the visible subset of patches (without mask tokens), along with a lightweight decoder that reconstructs the original image from the latent representation and mask tokens. MAE is pretrained by the ImageNet [15] dataset with a mask ratio of 75%. We include MAE as a comparison method to comprehensively enhance the robustness of our evaluation.

A. Scene Classification

Five downstream datasets shown in Table I are used to assess the performance of each pretrained model on scene classification task, including UC Merced (UCM) [56], Aerial image dataset (AID) [57], PatternNet [58], NWPU-RESISC45 (NR) [59], and EuroSAT (RGB version) [60]. The Top-1 accuracy serves as the performance metric.

1) Fine-Tuning Settings: For the scene classification downstream task, we add a linear classifier at the end of each pretrained model as the output stage. Note that the projector is maintained before the classifier in our PIS models. Following the settings of supervised training in TOV [42], we fine-tune these pretrained models using 5, 20, and 50 labeled samples per category, respectively. During PIS fine-tuning, we select Adam optimizer with an initial learning rate of $5e^{-4}$ and $2e^{-4}$, and a batch size of 12 and 4, for PIS-R50 and PIS-SwinB, respectively. For the other eight models, we adhere to the optimal configurations as mentioned in their original publications [20], [23], [33], [36], [42], [43]. The learning rate follows a cosine annealing schedule within 100 epochs. All the fine-tuning experiments are conducted on 1 NVIDIA GeForce RTX 3090 GPU.

2) Results and Analysis: The results are detailed in Table II. Notably, the two highest-performing results are marked in bold, with the leading result further highlighted in bold and red. Intuitively, our two PIS models occupy the top two positions no matter how many samples are used for fine-tuning, showcasing the effectiveness and strong generalizability of

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TABLE II
TOP-1 ACCURACY (%) OF SCENE CLASSIFICATION FOR FIVE DATASETS

Proportion of training data	Method	UCM	AID	PatternNet	NR	EuroSAT	Mean
5 samples	ResNet50 (Random)	30.39	23.84	35.41	22.77	22.01	26.88
	ResNet50(ImageNet)	51.42	46.67	79.20	41.47	39.45	51.64
	SwAV	74.52	66.48	86.28	**57.47**	61.96	69.34
	SeCo	69.32	57.21	82.70	48.07	61.13	63.69
	SSL4EO	68.86	53.57	79.15	44.78	59.34	61.14
	TOV	72.14	63.49	85.50	55.51	57.44	66.82
	GeRSP	**75.32**	**68.95**	86.34	56.69	38.70	65.20
	MAE	70.24	51.84	81.53	48.12	64.03	63.15
	PIS-R50	72.49	68.00	**87.97**	55.46	**66.15**	**70.01**
	PIS-SwinB	**75.00**	**69.18**	**89.98**	**60.68**	**78.40**	**74.65**
20 samples	ResNet50 (Random)	58.90	43.49	76.81	36.68	35.30	50.24
	ResNet50(ImageNet)	68.51	78.17	90.88	67.53	53.24	71.67
	SwAV	89.27	83.35	95.68	76.12	76.11	84.11
	SeCo	86.08	74.11	92.90	65.38	69.94	77.68
	SSL4EO	84.91	72.05	91.20	64.24	69.12	76.30
	TOV	**91.27**	83.53	95.84	74.59	73.40	83.73
	GeRSP	88.27	82.00	94.25	74.69	58.21	79.48
	MAE	88.74	83.95	94.20	74.41	**80.36**	84.33
	PIS-R50	91.07	**84.78**	**95.85**	**76.52**	80.30	**85.70**
	PIS-SwinB	**91.67**	**85.47**	**96.27**	**78.41**	**88.67**	**88.10**
50 samples	ResNet50 (Random)	75.74	63.30	89.05	56.30	46.03	66.08
	ResNet50(ImageNet)	83.70	84.28	94.81	80.66	55.97	79.88
	SwAV	95.36	89.83	97.42	**84.75**	81.16	89.70
	SeCo	89.14	83.35	96.22	75.02	76.73	84.09
	SSL4EO	89.42	82.96	94.70	72.20	74.80	82.82
	TOV	**96.69**	**90.44**	97.62	83.64	79.80	89.64
	GeRSP	94.75	88.56	96.96	82.00	67.01	85.86
	MAE	95.45	89.38	94.52	82.58	82.87	88.96
	PIS-R50	95.02	**90.44**	**97.91**	82.04	**83.93**	**89.87**
	PIS-SwinB	**98.38**	**91.39**	**98.27**	**86.19**	**92.95**	**93.44**

our foundation models. And both of them outperform all comparison methods in terms of the average accuracy of the five datasets, which demonstrates their SOTA capability in scene classification tasks. To be more specific, under the five-sample scenario, the PIS model with the same ResNet-50 backbone as the comparison methods exhibits an average performance rise of 3.19% over the best comparison method-TOV [42], while PIS with Swin Transformer-Base backbone further enhances this strength by an additional 4.64%.

Shift our focus to the comparison between the ImageNet-pretrained model and SSL-pretrained models. The former with a huge cost of human annotations apparently surpasses the randomly initialized model without surprise, but there remains a considerable gap for it to keep up with the latter (e.g., at least 2.94% when fine-tuning using 50 samples). We attribute this discrepancy to the domain gap between natural images and RSIs mentioned in Section I.

TABLE III
DATASETS FOR SEMANTIC SEGMENTATION EXPERIMENTS

Dataset	DLRSD	Potsdam	LoveDA
Number of categories	17	5	7
Number of samples	2,100	38	5,987
Spatial resolution (m)	0.3	0.05	0.3
Image sizes	256×256	6000×6000	1024×1024

B. Semantic Segmentation

To evaluate the generalizability of our proposed PIS models in the context of semantic segmentation downstream task, three datasets shown in Table III are employed: DLRSD [61], ISPRS Potsdam [62], and LoveDA dataset [63], and the mean

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intersection over union (MIoU) is utilized as the metric for performance evaluation. The processing procedures for the three datasets are as follows.

1) **DLRSD** dataset contains a total of 17 classes, with the pixels in each image being categorized according to the 17 predefined labels. Same as the UCM [56] archive, it has 2100 images, each with a resolution of 256 × 256 pixels. For the purpose of supervised training, each image is randomly cropped to 224 × 224.
2) **Potsdam** dataset contains 38 UAV images with a size of 6000 × 6000. In our experiment, the "2_Ortho_RGB.zip" and "5_Labels_all_noBoundary.zip" are required. Each image and its corresponding label are cropped into 256 × 256, resulting in a new dataset with 20 102 images, from which the training samples are randomly selected and further cropped into 224 × 224 as input.
3) **LoveDA** dataset is originally divided into three parts for training, validation, and testing. We combine these three subsets and form a new dataset comprising 67 056 images. Following a similar procedure to that used for the Potsdam dataset [62], the training samples are cropped to 224 × 224 as input.

1) Fine-Tuning Settings: As for the configurations of fine-tuning, we apply the UperNet [64] framework as the segmentation decoder, integrated with a backbone of ResNet-50 or Swin Transformer-Base. Consistent with the supervised downstream scenario outlined in TOV [42], we fine-tune these pretrained models using 0.5%, 1%, and 5% labeled samples of each entire dataset. We execute the supervised training for 100 epochs with an Adam optimizer and a batch size of eight across all the pretrained models. For SwAV, TOV, GeRSP, and MAE, we set the initial learning rate based on recommendations from their original publications [23], [36], [42], [43] while 0.003 for the rest of comparison models, $5e^{-4}$ and $2e^{-4}$ for our PIS-R50 and PIS-SwinB, respectively. A cosine annealing schedule is also included. All the fine-tuning experiments are conducted on 1 NVIDIA GeForce RTX 3090 GPU.

2) Results and Analysis: The final results are shown in Table IV. Similar to what is in the scene classification task, our two PIS models secure the top two places in terms of the average MIoU across the three datasets in the semantic segmentation task. More specifically, there is up to a 6.24% performance improvement observed between our PIS-SwinB and the best comparison method when fine-tuning with 50 samples. Note that the ImageNet-pretrained model does not significantly improve the performance over the randomly initialized one in semantic segmentation as it does in the scene classification task. The main reason is that it is pretrained using the task of supervised classification, which operates at a coarser granularity than pixel-wise semantic segmentation. This emphasizes the weak generalizability and inefficiency of the supervised pretraining method once again.

TABLE IV
MIOU (%) OF SEMANTIC SEGMENTATION FOR THREE DATASETS

Proportion of training data	Method	DLRSD	Potsdam	LoveDA	Mean
0.50%	ResNet50 (Random)	5.51	41.62	29.10	25.41
	ResNet50 (ImageNet)	6.55	44.27	31.55	27.46
	SwAV	9.78	49.51	37.57	32.29
	SeCo	**10.67**	47.56	34.18	30.80
	SSL4EO	3.98	42.51	32.58	26.36
	TOV	10.45	**53.03**	34.32	32.60
	GeRSP	10.19	50.37	31.90	30.82
	MAE	7.03	43.97	32.98	27.99
	PIS-R50	10.61	50.98	**38.37**	**33.32**
	PIS-SwinB	**12.49**	**57.37**	**42.22**	**37.36**
1%	ResNet50 (Random)	9.99	46.04	33.83	29.95
	ResNet50 (ImageNet)	10.49	48.58	36.13	31.73
	SwAV	13.40	**57.72**	39.68	36.93
	SeCo	13.92	52.20	38.44	34.85
	SSL4EO	9.07	48.49	34.55	30.70
	TOV	12.88	57.56	40.07	36.84
	GeRSP	10.19	50.37	31.90	30.82
	MAE	9.37	45.58	35.89	30.28
	PIS-R50	**16.10**	56.08	**41.24**	**37.81**
	PIS-SwinB	**16.94**	**63.09**	**46.18**	**42.07**
5%	ResNet50 (Random)	17.91	58.02	42.67	39.53
	ResNet50 (ImageNet)	18.76	58.82	42.96	40.18
	SwAV	26.38	66.37	45.62	46.12
	SeCo	24.92	64.20	45.17	44.76
	SSL4EO	18.78	58.94	43.23	40.32
	TOV	25.93	**66.74**	46.37	46.35
	GeRSP	10.19	50.37	31.90	30.82
	MAE	23.70	60.63	45.45	43.26
	PIS-R50	**26.68**	65.20	**48.06**	**46.65**
	PIS-SwinB	**35.15**	**69.88**	**52.73**	**52.59**

C. Change Detection

We then apply the pretrained models on a typical remote sensing downstream application: change detection, which aims at finding out the changed part between two spatially aligned aerial images captured at different timestamps. It can be seen as a two-class segmentation task, where the label "1" indicates change. The CDD dataset [65] contains four pairs of 1900 × 1000 RSIs and seven pairs of 4725 × 2200 RSIs collected from GE. [66] generates a new dataset by cropping them into 256 × 256 patches, and the number of samples for training, validation, and testing is 10 000, 3000, and 3000, respectively. We also combine these three subsets and follow the same procedure in LoveDA.

1) Fine-Tuning Settings: We adopt a SOTA framework in change detection-BIT [67] for the supervised training. For

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TABLE V
RESULTS OF CHANGE DETECTION FOR CDD DATASETS

Proportion of training data	Method	CDD		
		Precision (%)	Recall (%)	F1 (%)
1%	ResNet50 (Random)	20.98	1.39	2.21
	ResNet50 (ImageNet)	64.20	32.41	40.12
	SwAV	71.60	52.69	58.61
	SeCo	63.90	40.90	47.80
	SSL4EO	63.98	39.11	45.75
	TOV	66.33	42.34	48.57
	GeRSP	74.42	52.08	59.98
	MAE	66.55	50.67	54.94
	PIS-R50	**79.49**	**53.69**	**62.46**
	PIS-SwinB	**81.43**	**54.91**	**64.04**
5%	ResNet50 (Random)	52.93	21.06	27.60
	ResNet50 (ImageNet)	76.37	45.15	53.35
	SwAV	85.35	59.36	68.30
	SeCo	80.40	50.46	59.51
	SSL4EO	75.83	44.79	53.65
	TOV	78.05	48.18	56.50
	GeRSP	87.93	**73.83**	79.69
	MAE	78.79	64.10	69.48
	PIS-R50	**89.29**	73.72	**80.20**
	PIS-SwinB	**88.94**	**75.17**	**80.95**
10%	ResNet50 (Random)	66.02	40.57	47.20
	ResNet50 (ImageNet)	78.57	55.14	62.62
	SwAV	91.13	82.34	86.51
	SeCo	76.21	54.05	61.47
	SSL4EO	76.70	56.92	63.72
	TOV	76.85	54.82	61.80
	GeRSP	**91.99**	82.15	86.59
	MAE	85.25	67.65	74.31
	PIS-R50	91.72	**83.81**	**87.40**
	PIS-SwinB	**92.08**	**84.17**	**87.79**

SwAV, TOV, GeRSP, and MAE, we set the initial learning rate based on recommendations from their original publications [23], [36], [42], [43]. In the other comparison methods with ResNet50, we utilize an Adam optimizer with an initial learning rate of 0.003. Our PIS-R50 and PIS-SwinB are optimized by an Adam optimizer with an initial learning rate of $5e^{-4}$ and $2e^{-4}$, respectively. 1%, 5%, and 10% samples of the entire dataset are chosen for the 100-epoch training with a batch size of 8. All the fine-tuning experiments are conducted on 1 NVIDIA GeForce RTX 3090 GPU.

2) Results and Analysis: Table V shows the final results of Precision, Recall, and $F1$-score. Considering the $F1$-score, our PIS models also surpass the comparison methods in almost all cases aligning with expectations. Additionally, we can also see an advancement in precision and recall of PIS models over others. We further visualize some results of the CDD dataset obtained by each model using 10% labeled samples to fine-tune in Fig. 4. Intuitively, the detection results produced by our PIS models are more comprehensive than those from other pretrained models, and the missed detection areas shown within the red circle are much smaller.

D. Object Detection

Object detection is also a crucial downstream task in the interpretation of RSIs. As pointed in Section I, RSIs are captured from an aerial perspective and cover extensive spatial areas, leading to significant variations in scale and orientation among densely distributed objects. Consequently, oriented bounding box (OBB) detection is commonly employed in remote sensing. To further assess the generalizability of our foundation models, we choose the DOTA [68] dataset for our experiments. DOTA [68] is a large-scale dataset for aerial OBB detection, sourced from multiple sensors and satellites. It comprises a total of 2806 RSIs and 188 282 instances across 15 categories, with image sizes ranging from 800 × 800 to 20 000 × 20 000 pixels. These images feature objects of diverse scales, orientations, and shapes. The number of samples for training, validation and testing is 1411, 458, and 937, respectively. For the supervised training phase, we sample and crop the RSIs into 1024 × 1024 patches with a stride of 824. The data augmentations include random horizontal and vertical flipping.

1) Fine-Tuning Settings: We utilize ORCN [69], one of the SOTA OBB detection frameworks, to evaluate the performance of each pretrained model during the supervised training. For SwAV, TOV, GeRSP, and MAE, we set the initial learning rate according to recommendations found in their original publications [23], [36], [42], [43]. For other comparison methods with ResNet50, we use an Adam optimizer with an initial learning rate of $3e^{-3}$. Our PIS-R50 and PIS-SwinB are optimized by an Adam optimizer with an initial learning rate of $8e^{-3}$ and $1e^{-4}$, respectively. For convenience and better visualization results, we follow the standard configuration in [70]. The training and validation sets are merged for the 100-epoch training with a batch size of 4, while the testing set is used for evaluation. All the fine-tuning experiments are conducted on 1 NVIDIA GeForce RTX 3090 GPU.

2) Results and Analysis: Table VI presents the final results of the average precision (AP) of each category and the mean average precision (mAP) across all categories. Considering the AP per category, our PIS-R50 model outperforms the comparison methods in almost all cases, thereby achieving the highest mAP for ResNet-50 backbone. This underscores the superiority of our pretraining method against others after sufficient supervision. Additionally, our PIS-SwinB foundation model makes an extra step forward in AP per category, contributing to an improvement of 2.7% in mAP. We further visualize some detection results of the ORCN model with the randomly initialized, ImageNet-pretraining and our two versions of models on the DOTA testing dataset in Fig. 5. As we can see, the randomly initialized ResNet-50 missed both the densely distributed small vehicles (SV) in the red box and

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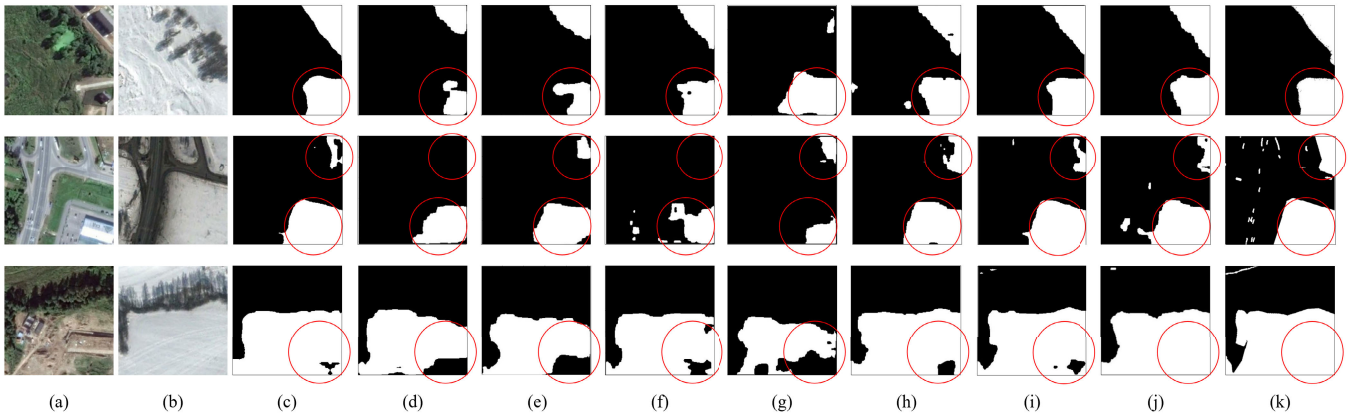


Fig. 4. Visualization results of each model on CDD dataset. (a) Original T0 images. (b) Original T1 images. (c) SwAV. (d) SeCo. (e) SSL4EO. (f) TOV. (g) GeRSP. (h) MAE. (i) PIS-R50. (j) PIS-SwinB. (k) Ground truth. Our two PIS models achieve better performances than other comparison methods.

TABLE VI
RESULTS OF OBJECT DETECTION FOR DOTA DATASETS

Method	AP per category (%)[1]															mAP (%)
	Plane	BD	BG	GTF	SV	LV	Ship	TC	BC	ST	SBF	RA	HB	SP	HC	
ResNet50 (Random)	78.36	47.20	24.18	23.56	56.86	58.25	69.73	69.99	58.88	68.21	23.46	41.24	33.87	45.91	28.21	48.53
ResNet50 (ImageNet)	85.24	61.28	45.72	62.46	71.20	73.92	81.47	90.12	73.52	79.90	44.35	55.22	59.87	55.92	31.54	64.78
SwAV	89.21	70.24	44.49	61.99	75.19	77.86	86.82	90.85	81.01	82.80	41.09	61.15	59.85	63.34	**58.14**	69.60
SeCo	88.63	70.03	44.59	51.90	71.84	75.86	85.84	90.77	73.72	80.17	43.35	57.17	62.71	57.50	38.55	66.18
SSL4EO	86.04	62.17	41.97	47.15	70.97	74.67	78.67	90.80	68.66	77.14	25.19	53.88	55.79	61.87	30.67	61.71
TOV	89.29	75.01	46.40	65.48	72.83	76.19	87.04	90.87	**83.27**	83.41	52.13	**66.20**	64.37	**68.46**	**51.80**	71.52
GeRSP	88.61	**78.33**	50.38	**66.21**	72.47	76.77	87.16	90.82	82.89	83.24	50.49	63.87	64.17	65.07	49.83	71.35
MAE	87.57	71.64	46.67	62.92	74.54	**78.78**	85.52	90.83	81.11	81.22	39.43	57.91	61.03	62.17	42.68	68.27
EMP-RS (ResNet50)	**89.53**	76.91	**50.42**	66.13	**77.41**	76.85	**87.55**	**90.90**	**84.94**	**83.80**	**52.26**	64.67	**65.09**	66.84	47.45	**72.05**
EMP-RS (Swin B)	**89.45**	**82.25**	**52.43**	**71.19**	**78.01**	**82.22**	**87.43**	**90.89**	83.09	**84.73**	**57.03**	**66.00**	**73.49**	**71.36**	51.63	**74.75**

[1] BD: baseball diamond. BG: bridge. GTF: ground track field. SV: small vehicle. LV: large vehicle. TC: tennis court. BC: basketball court. ST: storage tank. SBF: soccer ball field. RA: roundabout. HB: Harbor. SP: swimming pool. HC: helicopter.

large vehicles (LV) in the black box. Worse still, it mistakenly identifies LV as a plane, shown in the purple detection box adjacent to the black one. The ImageNet-pretraining model improves the detection for LV but still fails to detect the SV in the red box. In contrast, our PIS-R50 and PIS-SwinB accurately detect all the vehicles in the RSI without any mistake.

E. Ablation Study

To demonstrate the effectiveness of seasonal pairs and the increase in intra-instance variations, we carry out ablation studies on these two attributes. The two complete foundation models discussed in Section IV are hereby renamed as PIS-R50-IV16 + SP and PIS-SwinB-IV16 + SP for clarity, indicating the number of intra-instance variations (IV16) and the existence of seasonal pairs (+SP).

Initially, we pretrain two new foundation models, called PIS-R50-IV16 and PIS-SwinB-IV16, without seasonal pairs (e.g., only Spring timestamps are used) following the same training configurations as mentioned in IV. Then, we decrease the number of intra-instance variations to 4, i.e., the four timestamps are all used, and adjust the initial learning rate to 0.075 and $7.5e^{-5}$ in line with the linear learning rate scaling recipe [71] to obtain PIS-R50-IV4 + SP and PIS-SwinB-IV4 + SP, respectively. The performance of all PIS models is evaluated on the above ten datasets spanning four tasks, with the findings presented in Fig. 6.

Regardless of the backbone used, the models pretrained without seasonal pairs and with fewer intra-instance variations exhibit inferior performance compared to PIS-R50-IV16 + SP and PIS-SwinB-IV16 + SP in nearly all experiments across the four downstream tasks. This conclusively proves the effectiveness of incorporating seasonal pairs and increasing intra-instance variations.

F. Qualitative Analysis

We visualize the representations of the EuroSAT dataset extracted by each pretrained model, along with a randomly initialized model, using t-SNE [52]. The pretrained model is frozen and outputs a 128-dimension representation vector for each image, and each color stands for one of the ten categories

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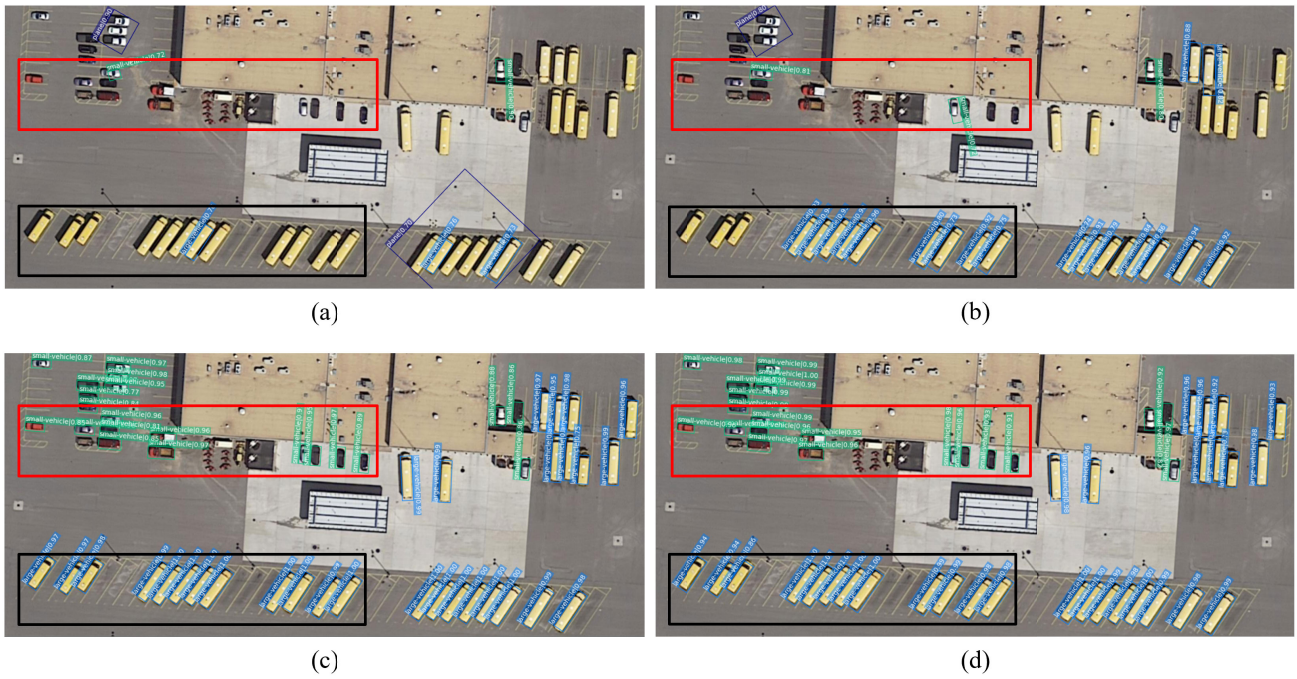


Fig. 5. Visualization detection results of the ORCN model with (a) randomly initialized, (b) ImageNet pretraining, (c) PIS-R50, and (d) PIS-SwinB backbones on the DOTA testing set.

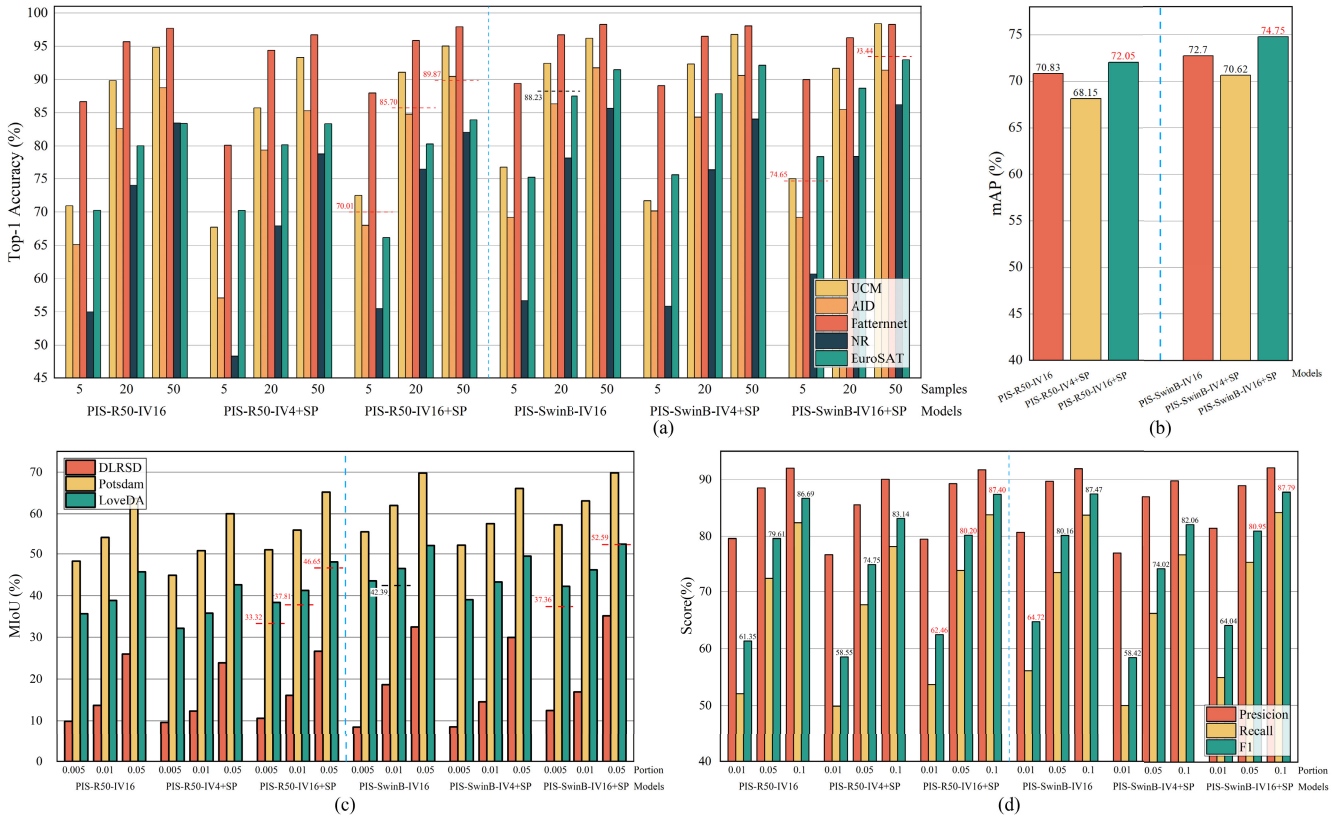


Fig. 6. Results of ablation studies. (a) Scene classification. (b) Object detection. (c) Semantic segmentation. (d) Change detection. The highest average performance in each scenario achieved by PIS-R50-IV16-E30-b128 + SP and PIS-SwinB-IV16-E30-b48 + SP is depicted as red lines in (a) and (c). Otherwise, it is shown as a black line. The highest mAP and $F1$-scores achieved by them are displayed in red digits in (b) and (d), respectively. Models with seasonal pairs and more intra-instance variations perform better.

within the EuroSAT dataset [60]. As shown in Fig. 7, our PIS-R50 and PIS-SwinB models learn much more separated and structured representations for different classes, demonstrating the exceptional capability of extracting latent features of RSIs.

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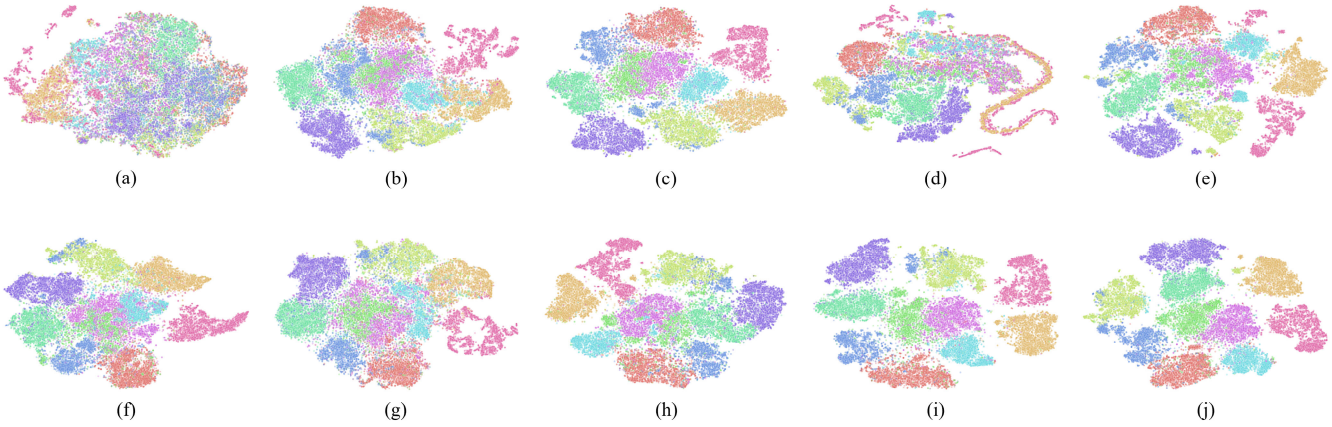


Fig. 7. t-SNE of learned representation on EuroSAT dataset for each pretrained model. (a) Random initialization. (b) ImageNet-pretrained. (c) SwAV. (d) SeCo. (e) SSL4EO. (f) TOV. (g) GeRSP. (h) MAE. (i) PIS-R50. (j) PIS-SwinB.

V. Conclusion

In this article, we introduce a remote sensing foundation model called PIS to supplant the common paradigm of loading ImageNet pretrained weights for supervised learning. Considering the long-standing issues of excessive attention to inter-instance discrimination and the suboptimal *two-variations-as-one-pair* pattern, we incorporate seasonal pairs and increase the number of intra-instance variations to expand the positive representation space, which alleviates the influence of false-negative and false-positive samples. Furthermore, to explore the intra-instance characteristics within this expanded space, we propose to promote intra-instance similarity for the purpose that the model can extract more instance-invariant features from different variations of the same instance. By fine-tuning the two pretrained models, PIS-R50 and PIS-SwinB, using a limited number of labeled samples, we achieve SOTA performances on ten datasets across four downstream tasks, demonstrating the generalizability and effectiveness of our proposed method.

Constrained by computational resources, the number of intra-instance variations is restricted to 16. Although the PIS models have achieved SOTA performances even under this condition, we anticipate that their capabilities can be fully brought out when pushing the number of intra-instance variations to an extreme as suggested by our ablation study on this factor. Additionally, while our PIS models can still achieve excellent performance without seasonal pairs, as demonstrated by the ablation studies, their full potential is realized with the inclusion of seasonal pairs. Thanks to the SSL4EO-S12 dataset, which provides us with four seasonal timestamps as extra natural augmentations, we are allowed to expand the positive representation space. However, our method is limited in locations where seasonal pairs are unavailable. In such cases, we would need to invest extra effort to register the unorganized images, which hinders the scalability of our method. An alternative approach could involve using diffusion models to generate more intrinsic intra-instance variations, thereby eliminating the need for registration. Our future research will explore the feasibility of this method.

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